

The Vaccination Coverage Needed To Reduce An Infectious Disease Outbreak

Introduction

When many individuals in a population are at risk of an infectious disease, it can spread rapidly. Decisions about controlling an outbreak are often made before the entire outbreak pattern is known, which causes serious issues for public health. Mechanistic models are useful due to how they can provide potential tested assumptions about transmission of the disease, recovery and possible ways to control, although they cannot thoroughly predict a real outbreak, but can show generalised patterns to support strategic planning in these situations.

One approach is the susceptible-infectious-recovered model (SIR model). This breaks the population into 1 of 3 groups: individuals who can catch the disease, individuals with the disease and can pass it on, and individuals who have recovered or are immune to the disease (Hethcote, 2000). In an SIR model, the reproduction number, R_0 , is the average number of secondary cases caused by one infectious person in a fully susceptible population. If the R_0 is above 1, it means the infection can spread, and if the R_0 is below 1, then the infection should decrease (Heffernan et al., 2005).

Vaccinations are a preventative measure to reduce the spread of a disease by lowering the number of susceptible individuals. Vaccines directly protect individuals as well as reduce transmission in the whole population through herd immunity (Fine et al., 2011). Vaccinations have been crucial for controlling the spread of many infectious diseases (Anderson and May, 1982). Even so, if the vaccine coverage is too low, an outbreak might still take place. This uses an SIR model to research how vaccination coverage before an outbreak affects disease spread. The aim was to see how vaccination coverage changes the time until peak and total infected during an outbreak as well as the peak active infections through an outbreak. It was predicted that higher vaccination coverage would reduce the size of the outbreak as well as reduce the spread of the disease, the closer to the theoretical herd immunity threshold.

Methods

A susceptible infectious recovered model was used to simulate the spread of an infectious disease, with the population split into 3 groups: susceptible, infectious and recovered / immune. Susceptible individuals could catch the disease, infectious individuals could transmit it, and recovered / immune individuals were unable to catch the disease.

The population was proportional, so the total population could always be equal to 1. It was assumed that susceptible individuals were infected through contact with infectious individuals, with the speed of infection depending on the transmission rate, β . The

infectious individuals then transitioned into the recovered / immune group, depending on the recovery rate, gamma.

The reproduction number, R_0 , was set to 3. This meant that for every infectious individual would infect 3 others on average in a fully susceptible population, with the recovery rate set to $1/7$, meaning that on average, individuals were infectious for 7 days. The transmission rate was then calculated using R_0 and the recovery rate.

Vaccinations were included by putting individuals into the recovered / immune group before the outbreak started, which reduces the number of susceptible individuals at the beginning of the outbreak. The initial infected proportion was set to 0.001 (0.1% of the population). To start, the model was run at 0%, 30%, 50%, 66.7%, 80% and 90% vaccination coverage. The threshold for theoretical herd immunity at R_0 of 3 is 66.7% and is therefore why it was included. The model was then also run at incremental 1% increases between 0% to 95% vaccination coverage to get a broader pattern.

The simulations were run using the desolve package in R Studio, and the graphs were made using ggplot2 in R Studio. This gave the peak active infections, time until peak infected and the total infected during the outbreak. The total infected through the outbreak was calculated by deducting the number vaccinated at the beginning of the outbreak from the number recovered at the end.

Results

The outbreak size and the peak of active infections both declined when vaccines were introduced. The model produced a widespread outbreak without vaccination. Active infections peaked at 30.1% of the population after 26.8 days, with 94.1% of the overall population becoming infected.

When vaccination coverage was at 30%, the outbreak was smaller, but it was still widely spread, with the initial effective reproduction number decreased to 2.10, peak active infections went down to 12%, with 57.6% of the population became infected. When there was a 50% vaccination rate, peak active infections went down to 3.22%, and the total population infected during the outbreak decreased to 29%, with the peak taking longer at 69.7 days.

In the 66.7% vaccination rate simulation, active infections did not rise above the initial 0.1% starting level, and the total infected throughout the outbreak was decreased to 1.79%. This indicates the outbreak was almost stopped at the theoretical threshold. When going above the threshold, the infectious disease did not spread successfully, at 80%, the total infections fell to 0.25% and fell even further to 0.14% at 90% vaccination coverage.

The height of the infection curve decreases as the vaccination coverage increases, as seen in Figure 1. Higher vaccination coverage also delayed the peak when an outbreak did still happen. Figure 2 shows that both the total outbreak size and the peak active infections decreased as vaccination coverage increased, with widespread outbreaks disappearing close to the 66.7% theoretical threshold.

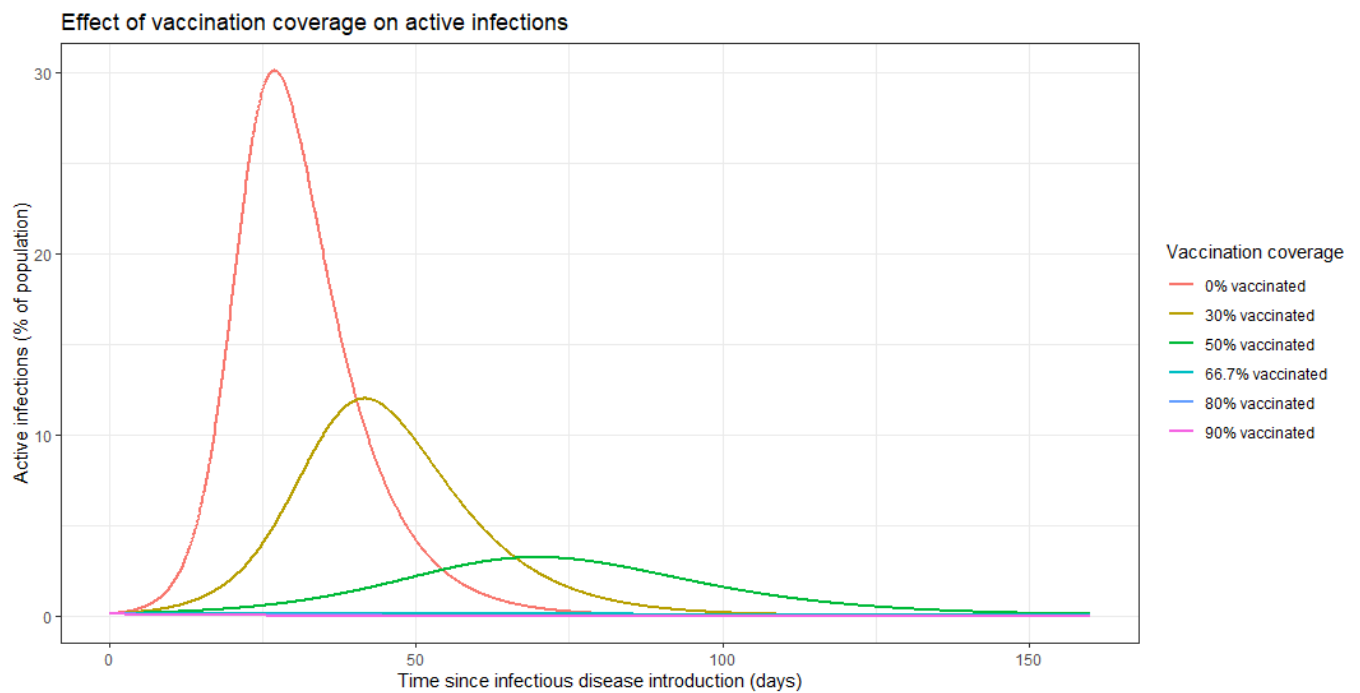


Figure 1: active infections over time with 6 different vaccination coverage levels. Vaccination was before infectious disease was added by moving the vaccinated individuals to the recovered / immune group.

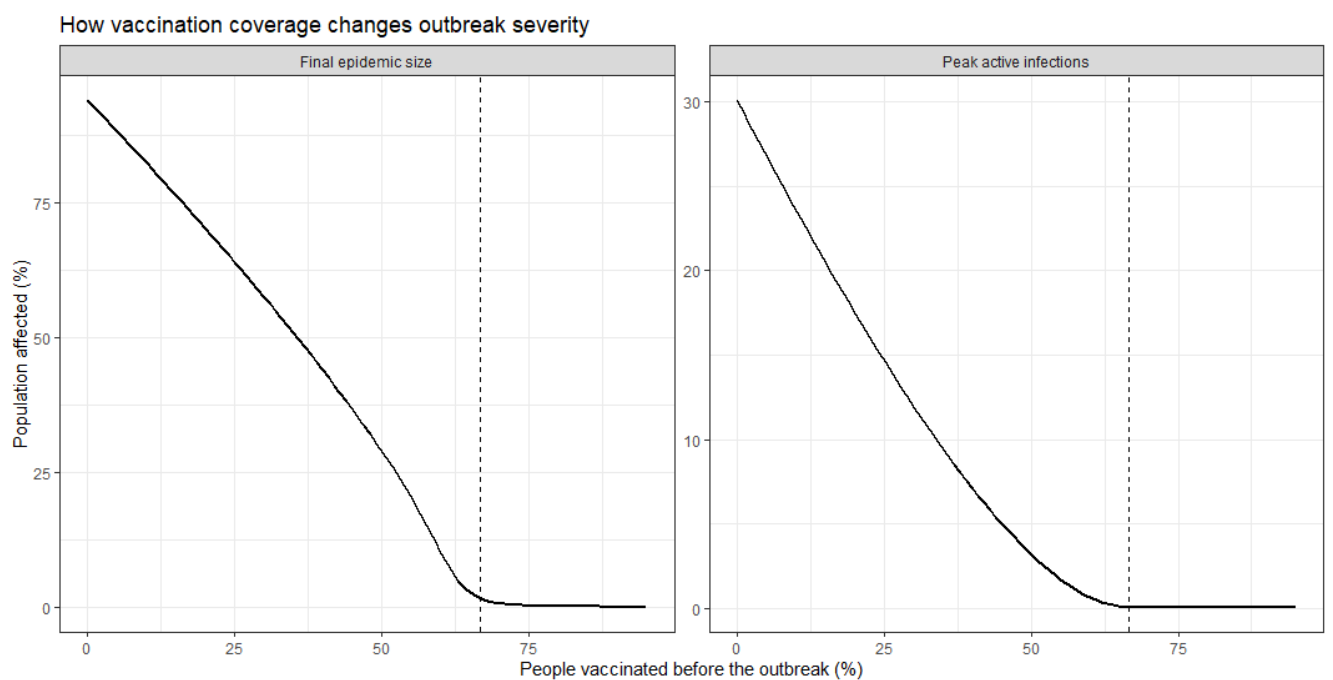


Figure 2: The effect of vaccination coverage on the total infected individuals in the outbreak and peak active infections. 66.7% is marked with a line to show the theoretical threshold for herd immunity.

Discussion

These results support the hypothesis that increasing the vaccination coverage will reduce the spread of the disease. With no vaccination, nearly the whole population became infected due to the initial effective reproduction number being so high. As starting vaccination coverage increased, there were fewer people susceptible at the beginning and therefore fewer opportunities for the infectious disease to spread.

The model showed a clear threshold effect, when the vaccination coverage is below 66.7%, outbreaks still happened, but were smaller as the vaccination coverage increased. The outbreak was greatly reduced at 50% compared to no vaccine, but 29% of the population was still infected. This shows that the infectious disease might slow and be less effective with partial vaccination, but it isn't enough to stop transmission. Active infections only didn't rise above the initial level at 66.7%, which matches the theoretical herd immunity, and the idea that the susceptible population is reduced enough that disease spread is prevented (Fine et al., 2011).

These results are consistent with previous research showing that vaccinations can control transmitted diseases (Anderson and May, 1985). Maintaining high vaccination coverage is important, and if the coverage falls below the threshold, then a population might be vulnerable to an outbreak of an infectious disease. Disruption to vaccination plans can increase disease burden in the future (Gaythorpe et al., 2021).

There are many limitations to this model, including the assumption that all individuals are mixed equally, unlike in a real population, as well as contact rates staying at a constant across the whole population, which may differ with age, location and behaviour. Real vaccines are not always 100% effective, and the model doesn't include birth, death and repeated introductions of the infectious disease. The model is also deterministic and therefore does not include random chance, which is important when outbreaks are small.

Future work could make the model more realistic, including imperfect vaccine protection and different R_0 values. Even taking that into account, the SIR model shows the importance of vaccination coverage for reducing outbreak size.

Conclusion

There was a strong decline in outbreak size and peak infections with increasing vaccination coverage, with most disease spread being prevented close to the theoretical threshold at 66.7%. This shows the importance of high vaccination coverage to reduce the effective reproduction number and prevent transmutation.

References

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Gaythorpe, K.A., K. Abbas, J. Huber, A. Karachaliou, N. Thakkar, K. Woodruff, X. Li, S. Echeverria-Londono, M. Ferrari, M.L. Jackson, K. McCarthy, T.A. Perkins, C. Trotter, and M. Jit. 2021. Impact of COVID-19-related disruptions to measles, meningococcal A, and yellow fever vaccination in 10 countries. *eLife*. 10:e67023. doi:<https://doi.org/10.7554/eLife.67023>.

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Appendix

```
install.packages("deSolve")
install.packages("ggplot2")
install.packages("dplyr")
install.packages("tidyr")
library(deSolve)
library(ggplot2)
library(dplyr)
library(tidyr)
#parameters
R0_basic <- 3
gamma <- 1 / 7
beta <- R0_basic * gamma
params_sir <- c(beta = beta, gamma = gamma)
initial_infected <- 0.001
time <- seq(0, 160, by = 0.1)
herd_threshold <- 1 - (1 / R0_basic)
herd_threshold
#herd threshold 0.6666667

#Sir(model)prize
model_sir <- function(t, y, params) {
  with(as.list(c(params, y)), {
    dS <- -beta * S * I
```

```

dI <- beta * S * I - gamma * I
dR <- gamma * I
list(c(dS, dI, dR))
})
}
# one vaccination coverage
simulate_sir <- function(vaccination_coverage) {
  init_sir <- c(
    S = 1 - initial_infected - vaccination_coverage,
    I = initial_infected,
    R = vaccination_coverage
  )
  out_sir <- as.data.frame(
    ode(
      y = init_sir,
      times = time,
      func = model_sir,
      parms = params_sir
    )
  )
  out_sir$vaccination_coverage <- vaccination_coverage
  out_sir$vaccination_percent <- vaccination_coverage * 100

  return(out_sir)
}
#vaccination with dif cover
selected_vaccination <- c(0, 0.30, 0.50, herd_threshold, 0.80, 0.90)
selected_models <- lapply(selected_vaccination, simulate_sir)
sir_curves <- bind_rows(selected_models)
sir_curves$vaccination_label <- paste0(
  round(sir_curves$vaccination_coverage * 100, 1),
  "% vaccinated"
)

```

```

#graph time for figure
figure_1 <- ggplot(
  sir_curves,
  aes(
    x = time,
    y = I * 100,
    colour = vaccination_label
  )
) +
  geom_line(linewidth = 1) +
  labs(
    title = "Effect of vaccination coverage on active infections",
    x = "Time since infectious disease introduction (days)",
    y = "Active infections (% of population)",
    colour = "Vaccination coverage"
  ) +
  theme_bw()
figure_1

# outbreaks across vaccination cover
vaccination_gradient <- seq(0, 0.95, by = 0.01)
all_models <- lapply(vaccination_gradient, simulate_sir)
all_outputs <- bind_rows(all_models)
#measure outbreak size, time and infections
summary_results <- all_outputs %>%
  group_by(vaccination_coverage) %>%
  summarise(
    vaccination_percent = first(vaccination_percent),
    initial_Re = R0_basic * first(S),
    peak_infectious = max(I),
    time_to_peak = time[which.max(I)],
    final_recovered = last(R),
    final_epidemic_size = final_recovered - first(vaccination_coverage),
  )

```

```

.groups = "drop"
)
summary_results <- summary_results %>%
mutate(
  Peak_active_infections = peak_infectious * 100,
  final_epidemic_size_percent = final_epidemic_size * 100
)
head(summary_results)

#another figure
summary_long <- summary_results %>%
select(
  vaccination_percent,
  final_epidemic_size_percent,
  Peak_active_infections
) %>%
pivot_longer(
  cols = c(final_epidemic_size_percent, Peak_active_infections),
  names_to = "outcome",
  values_to = "value"
) %>%
mutate(
  outcome = recode(
    outcome,
    final_epidemic_size_percent = "Final epidemic size",
    Peak_active_infections = "Peak active infections"
  )
)

#ggplot time.. finally
figure_2 <- ggplot(
  summary_long,
  aes(
    x = vaccination_percent,

```

```

    y = value
  )
) +
  geom_line(linewidth = 1) +
  geom_vline(
    xintercept = herd_threshold * 100,
    linetype = "dashed"
  ) +
  facet_wrap(~ outcome, scales = "free_y") +
  labs(
    title = "How vaccination coverage changes outbreak severity",
    x = "People vaccinated before the outbreak (%)",
    y = "Population affected (%)"
  ) +
  theme_bw()
figure_2

```

#results section

```

selected_outputs <- sir_curves %>%
  group_by(vaccination_coverage) %>%
  summarise(
    vaccination_percent = first(vaccination_percent),
    initial_Re = R0_basic * first(S),
    peak_active_infections = max(I),
    time_to_peak = time[which.max(I)],
    final_recovered = last(R),
    total_infected_during_outbreak = final_recovered - first(vaccination_coverage),
    .groups = "drop"
  ) %>%
  mutate(
    peak_active_infections_percent = peak_active_infections * 100,
    total_infected_during_outbreak_percent = total_infected_during_outbreak * 100
  )

```

```
selected_outputs_report <- selected_outputs %>%  
transmute(  
  vaccination_percent = round(vaccination_percent, 1),  
  initial_Re = round(initial_Re, 2),  
  peak_active_infections_percent = round(peak_active_infections_percent, 2),  
  time_to_peak_days = round(time_to_peak, 1),  
  total_infected_during_outbreak_percent = round(total_infected_during_outbreak_percent, 2)  
)  
  
selected_outputs_report  
print(selected_outputs_report, width = Inf)
```